

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

A quadratic function models a projectile's height, in meters, above the ground in terms of the time, in seconds, after it was launched. The model estimates that the projectile was launched from an initial height of **7** meters above the ground and reached a maximum height of **51.1** meters above the ground **3** seconds after the launch. How many seconds after the launch does the model estimate that the projectile will return to a height of **7** meters?

Answer

- A. **3**
- B. **6**
- C. **7**
- D. **9**

Correct Answer: B

Rationale

Choice B is correct. It's given that a quadratic function models the projectile's height, in meters, above the ground in terms of the time, in seconds, after it was launched. It follows that an equation representing the model can be written in the form $f(x) = a(x - h)^2 + k$, where $f(x)$ is the projectile's estimated height above the ground, in meters, x seconds after the launch, a is a constant, and k is the maximum height above the ground, in meters, the model estimates the projectile reached h seconds after the launch. It's given that the model estimates the projectile reached a maximum height of **51.1** meters above the ground **3** seconds after the launch. Therefore, $k = 51.1$ and $h = 3$. Substituting **51.1** for k and **3** for h in the equation $f(x) = a(x - h)^2 + k$ yields $f(x) = a(x - 3)^2 + 51.1$. It's also given that the model estimates that the projectile was launched from an initial height of **7** meters above the ground. Therefore, when $x = 0$, $f(x) = 7$. Substituting **0** for x and **7** for $f(x)$ in the equation $f(x) = a(x - 3)^2 + 51.1$ yields $7 = a(0 - 3)^2 + 51.1$, or $7 = 9a + 51.1$. Subtracting **51.1** from both sides of this equation yields $-44.1 = 9a$. Dividing both sides of this equation by **9** yields $-4.9 = a$. Substituting -4.9 for a in the equation $f(x) = a(x - 3)^2 + 51.1$ yields $f(x) = -4.9(x - 3)^2 + 51.1$. Therefore, the equation $f(x) = -4.9(x - 3)^2 + 51.1$ models the projectile's height, in meters, above the ground x seconds after it was launched. The number of seconds after the launch that the model estimates that the projectile will return to a height of **7** meters is the value of x when $f(x) = 7$. Substituting **7** for $f(x)$ in $f(x) = -4.9(x - 3)^2 + 51.1$ yields $7 = -4.9(x - 3)^2 + 51.1$. Subtracting **51.1** from both sides of this equation yields $-44.1 = -4.9(x - 3)^2$. Dividing both sides of this equation by -4.9 yields $9 = (x - 3)^2$. Taking the square root of both sides of this equation yields two equations: $3 = x - 3$ and $-3 = x - 3$. Adding **3** to both sides of the equation $3 = x - 3$ yields $6 = x$. Adding **3** to both sides of the equation $-3 = x - 3$ yields $0 = x$. Since **0** seconds after the launch represents the time at which the projectile was launched, **6** must be the number of seconds the model estimates that the projectile will return to a height of **7** meters.

Alternate approach: It's given that a quadratic function models the projectile's height, in meters, above the ground in terms of the time, in seconds, after it was launched. It's also given that the model estimates that the projectile was launched from an initial height of **7** meters above the ground and reached a maximum height of **51.1** meters above the ground **3** seconds after the launch. Since the model is quadratic, and quadratic functions are symmetric, the model estimates that for any given height less than the maximum height, the time the projectile takes to travel from the given height to the maximum height is the same as the time the projectile takes to travel from the maximum height back to the given height. Thus, since the model estimates the projectile took **3** seconds to travel from **7** meters above the ground to its maximum height of **51.1** meters above the ground, the model also estimates the projectile will take **3** more seconds to travel from its maximum height of **51.1** meters above the ground back to **7** meters above the ground. Thus, the model estimates that the projectile will return to a height of **7** meters **3** seconds after it reaches its maximum height, which is **6** seconds after the launch.

Choice A is incorrect. The model estimates that **3** seconds after the launch, the projectile reached a height of **51.1** meters, not **7** meters.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.